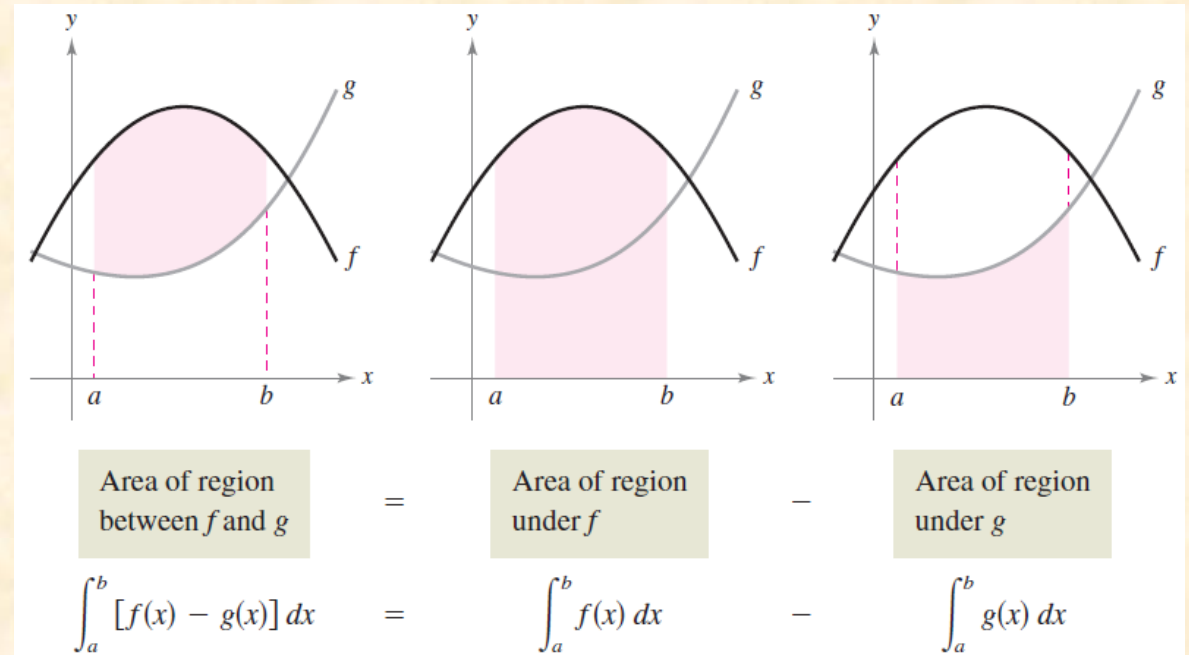
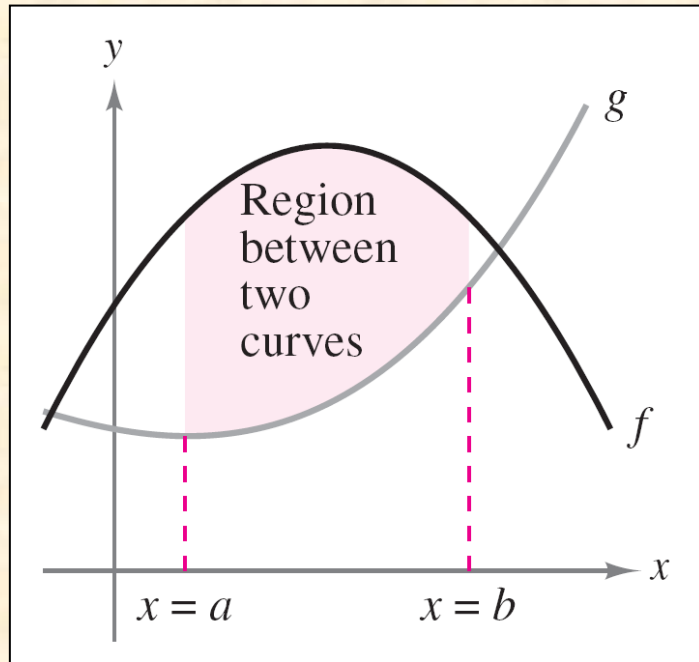


# **7.1: Area of a Region Between Two Curves**

# Area of a Region Between Two Curves



## Area of a Region Between Two Curves

If  $f$  and  $g$  are continuous on  $[a, b]$  and  $g(x) \leq f(x)$  for all  $x$  in  $[a, b]$ , then the area of the region bounded by the graphs of  $f$  and  $g$  and the vertical lines  $x = a$  and  $x = b$  is

$$A = \int_a^b [f(x) - g(x)] dx.$$

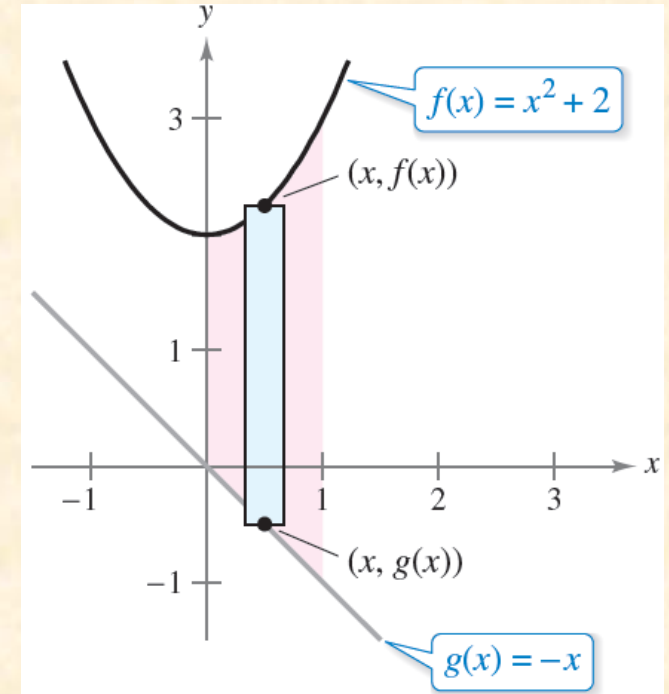
# Example 1 – Finding the Area of a Region Between Two Curves

Find the area of the region bounded by the graphs of  $y = x^2 + 2$ ,  $y = -x$ ,  $x = 0$ , and  $x = 1$ .

**Solution:**

Let  $g(x) = -x$  and  $f(x) = x^2 + 2$ .

Then  $g(x) \leq f(x)$  for all  $x$  in  $[0, 1]$ , as shown in Figure 7.5.



Region bounded by the graph of  $f$ , the graph of  $g$ ,  $x = 0$ , and  $x = 1$

**Figure 7.5**

# Example 1 – Finding the Area of a Region Between Two Curves

So, the area of the representative rectangle is

$$\Delta A = [f(x) - g(x)]\Delta x$$

$$= [(x^2 + 2) - (-x)]\Delta x$$

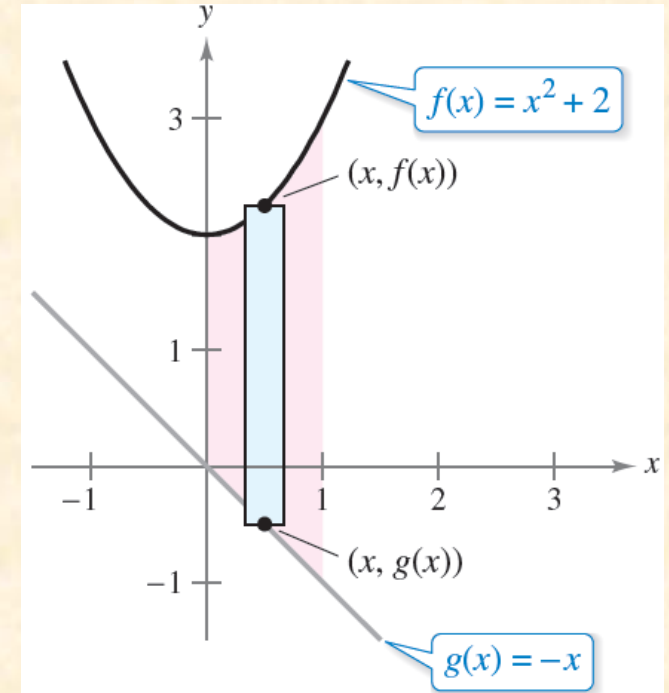
and the area of the region is

$$A = \int_a^b [f(x) - g(x)] dx$$

$$= \int_0^1 [(x^2 + 2) - (-x)] dx$$

$$= \left[ \frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_0^1$$

$$= \frac{1}{3} + \frac{1}{2} + 2 = \frac{17}{6}.$$



Region bounded by the graph of  $f$ , the graph of  $g$ ,  $x = 0$ , and  $x = 1$

**Figure 7.5**

# Area of a Region Between Intersecting Curves

## Example 2 – A Region Lying Between Two Intersecting Graphs

Find the area of the region bounded by the graphs of  $f(x) = 2 - x^2$  and  $g(x) = x$ .

### Solution:

In Figure 7.6, notice that the graphs of  $f$  and  $g$  have two points of intersection.

To find the  $x$ -coordinates of these points, set  $f(x)$  and  $g(x)$  equal to each other and solve for  $x$ .

$$2 - x^2 = x$$

$$-x^2 - x + 2 = 0$$

$$-(x + 2)(x - 1) = 0$$

$$x = -2 \text{ or } 1$$

So,  $a = -2$  and  $b = 1$ .

Set  $f(x)$  equal to  $g(x)$ .

Write in general form.

Factor.

Solve for  $x$ .

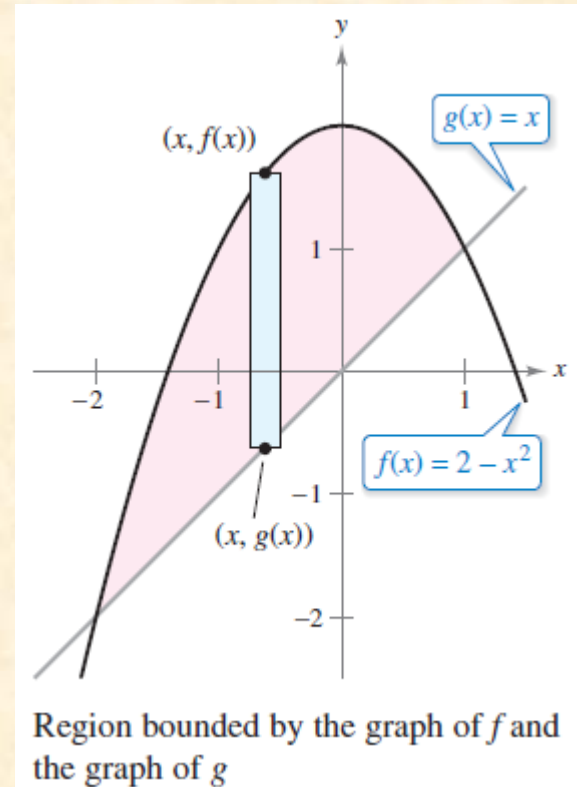


Figure 7.6

# Area of a Region Between Intersecting Curves

Because  $g(x) \leq f(x)$  for all  $x$  in the interval  $[-2, 1]$ , the representative rectangle has an area of

$$\begin{aligned}\Delta A &= [f(x) - g(x)]\Delta x \\ &= [(2 - x^2) - x]\Delta x\end{aligned}$$

and the area of the region is

$$A = \int_{-2}^1 [(2 - x^2) - x] dx$$

$$= \left[ -\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_{-2}^1$$

$$= \frac{9}{2}.$$

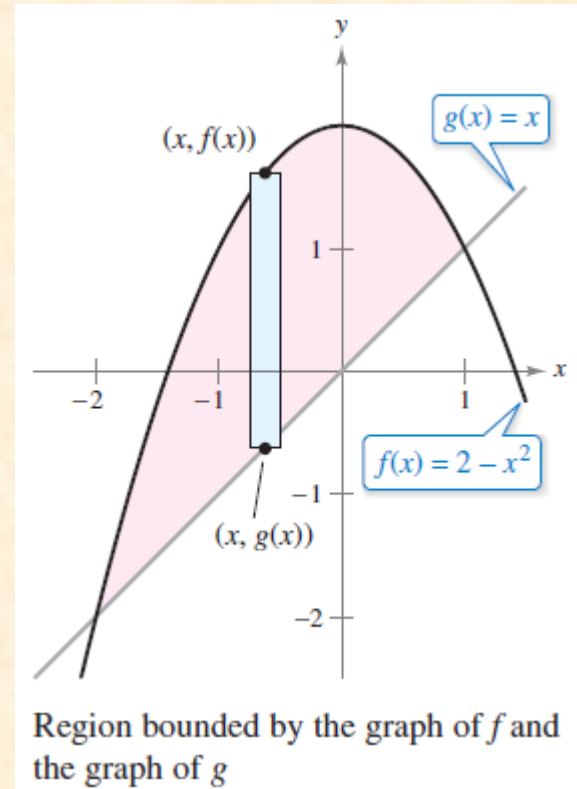


Figure 7.6

**7.2**

## Volume: The Disk Method

# Volume: The Disk Method

- When a region in the plane is revolved about a line, the resulting solid is a **solid of revolution**, and the line is called the **axis of revolution**.
- The simplest such solid is a right circular cylinder or **disk**, which is formed by revolving a rectangle about an axis adjacent to one side of the rectangle, as shown in Figure 7.13.
- The volume of such a disk is  
= (area of disk)(width of disk)  
Volume of disk =  $\pi R^2 w$   
where  $R$  is the radius of the disk and  $w$  is the width.

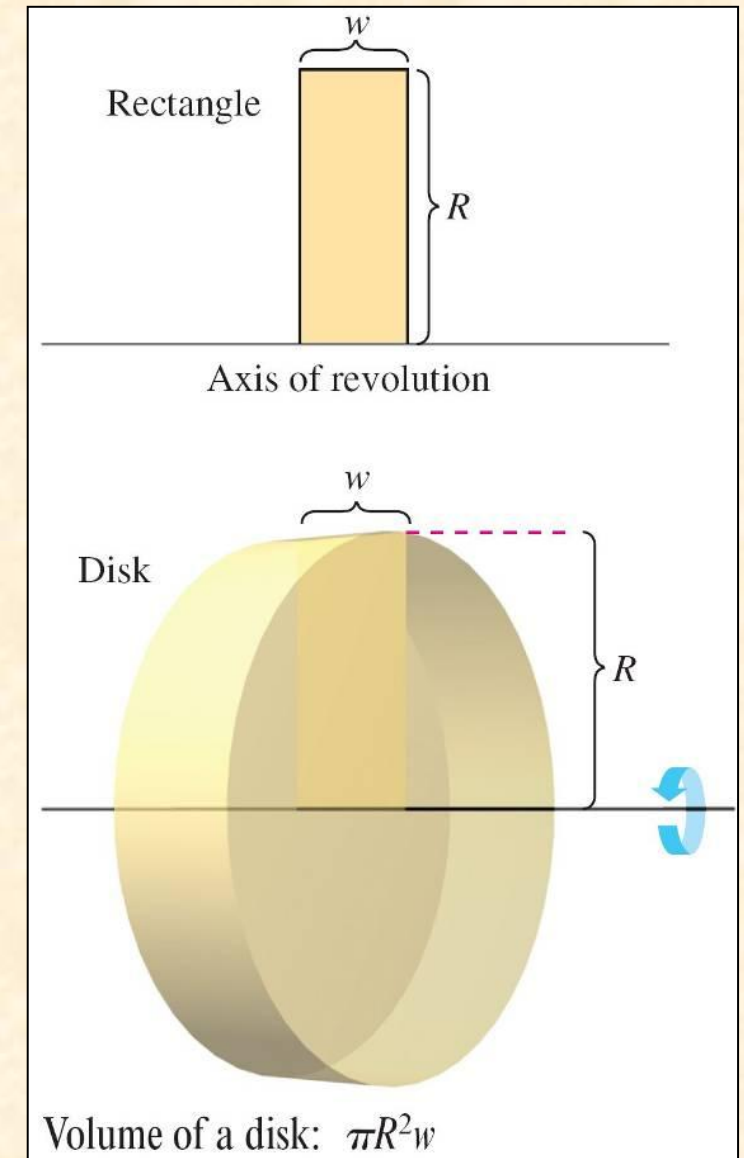


Figure 7.13

# Volume: The Disk Method

To see how to use the volume of a disk to find the volume of a general solid of revolution, consider a solid of revolution formed by revolving the plane region in Figure 7.14 about the indicated axis.

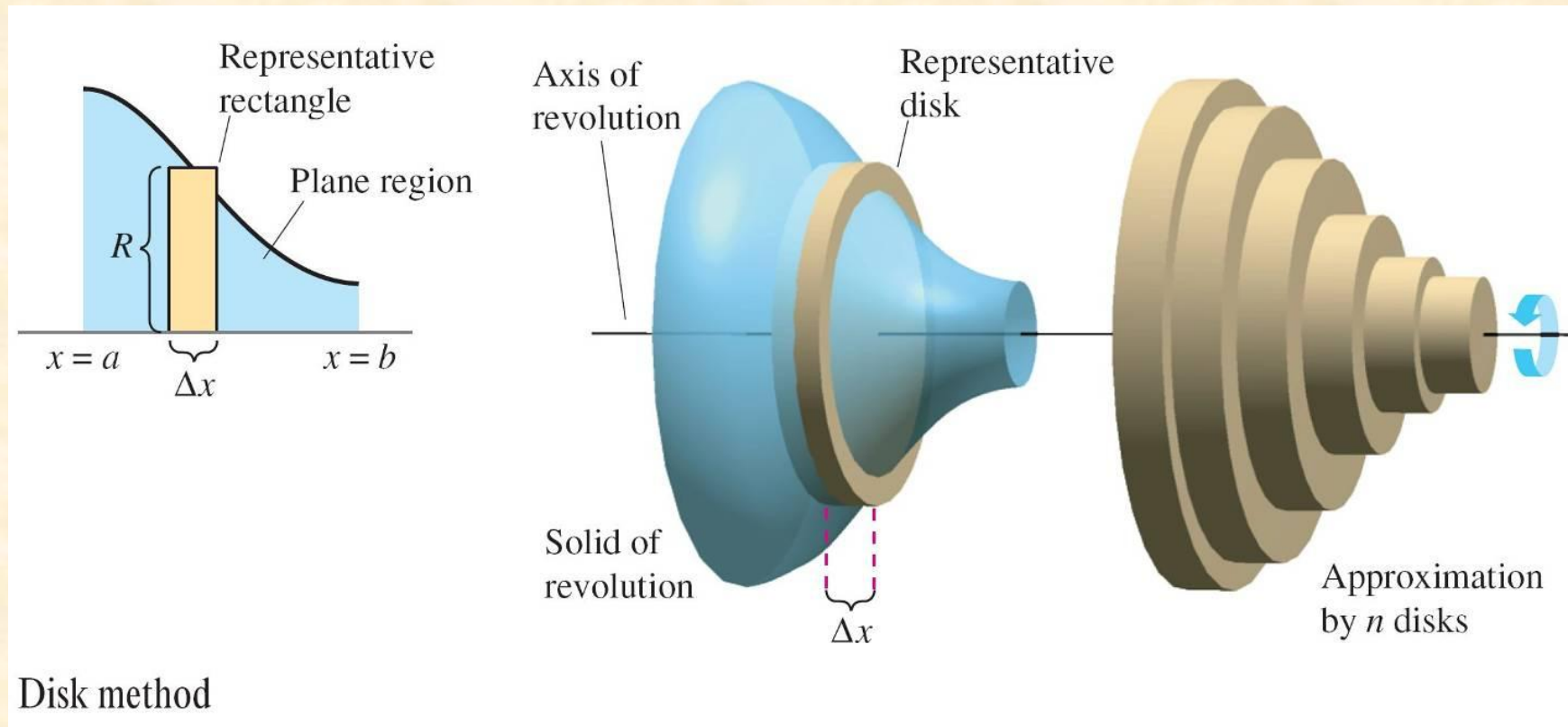


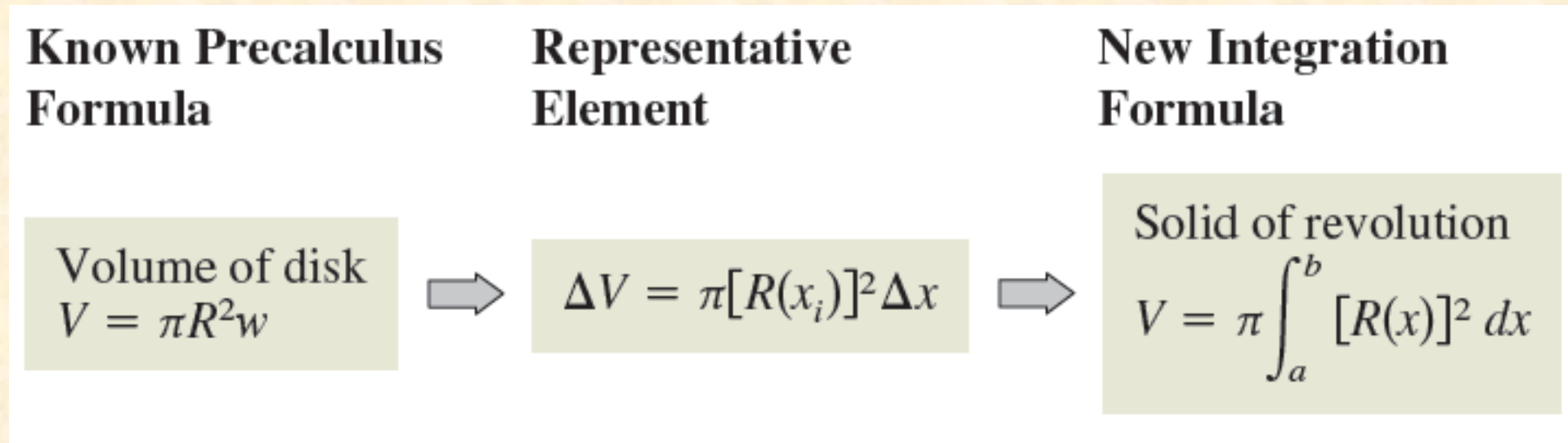
Figure 7.14

# Volume: The Disk Method

This approximation appears to become better and better as  $\|\Delta\| \rightarrow 0$  ( $n \rightarrow \infty$ ). So, you can define the volume of the solid as

$$\text{Volume of solid} = \lim_{\|\Delta\| \rightarrow 0} \pi \sum_{i=1}^n [R(x_i)]^2 \Delta x = \pi \int_a^b [R(x)]^2 dx.$$

Schematically, the disk method looks like this.



# Volume: The Disk Method

A similar formula can be derived when the axis of revolution is vertical.

## THE DISK METHOD

To find the volume of a solid of revolution with the **disk method**, use one of the formulas below. (See Figure 7.15.)

### Horizontal Axis of Revolution

$$\text{Volume} = V = \pi \int_a^b [R(x)]^2 dx$$

### Vertical Axis of Revolution

$$\text{Volume} = V = \pi \int_c^d [R(y)]^2 dy$$

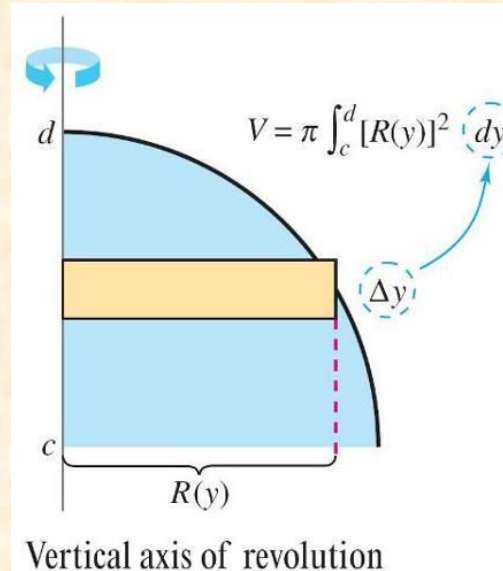
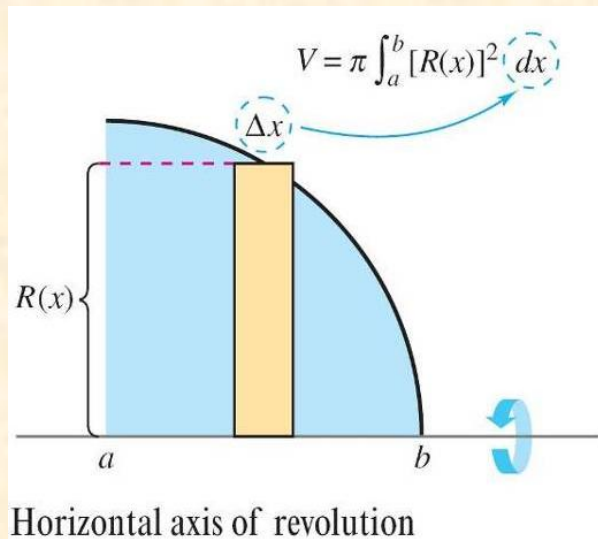


Figure 7.15

## Example 1 – Using the Disk Method

Find the volume of the solid formed by revolving the region bounded by the graph of  $f(x) = \sqrt{\sin x}$  and the  $x$ -axis ( $0 \leq x \leq \pi$ ) about the  $x$ -axis, as shown in Figure 7.16.

### Solution:

From the representative rectangle in the upper graph in Figure 7.16, you can see that the radius of this solid is

$$\begin{aligned} R(x) &= f(x) \\ &= \sqrt{\sin x}. \end{aligned}$$

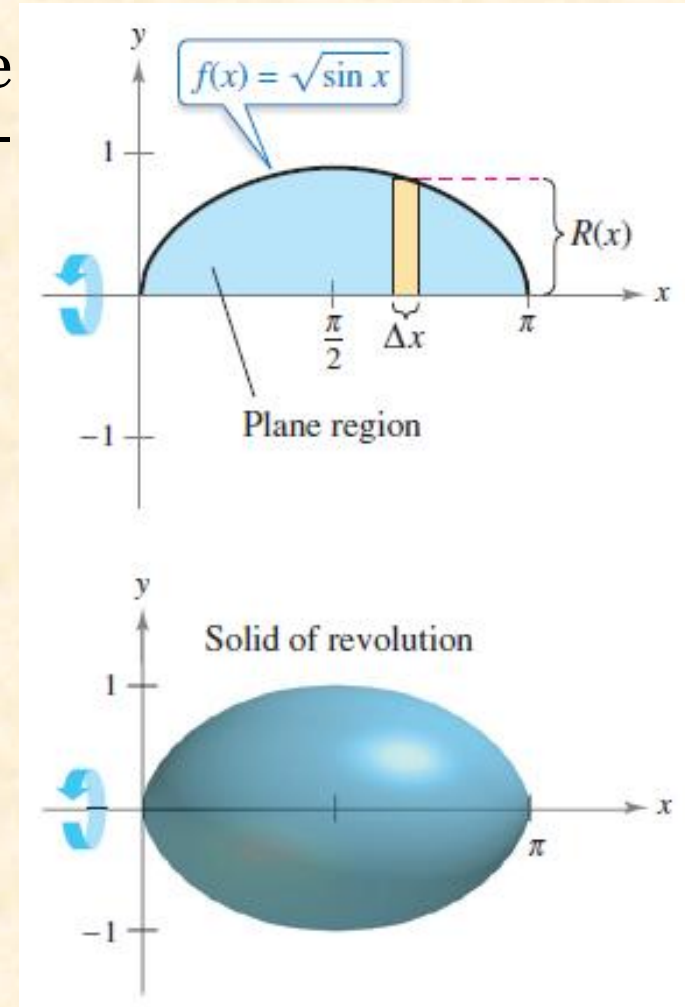


Figure 7.16

# Example 1 – Solution

So, the volume of the solid of revolution is

$$V = \pi \int_a^b [R(x)]^2 dx$$

Apply disk method.

$$= \pi \int_0^{\pi} (\sqrt{\sin x})^2 dx$$

Substitute  $\sqrt{\sin x}$  for  $R(x)$ .

$$= \pi \int_0^{\pi} \sin x dx$$

Simplify.

$$= \pi \left[ -\cos x \right]_0^{\pi}$$

Integrate.

$$= \pi(1 + 1)$$

$$= 2\pi.$$

# Try Yourself

1. Two models  $R_1$  and  $R_2$  are given for revenue (in millions of dollars) for a corporation. Both models are estimates of revenues from 2020 through 2025, with  $t = 0$  corresponding to 2020. Which model projects the greater revenue? How much more total revenue does that model project over the six-year period?

$$R_1 = 7.21 + 0.26t + 0.02t^2$$

$$R_2 = 7.21 + 0.1t + 0.01t^2$$

# The Washer Method

# The Washer Method

The disk method can be extended to cover solids of revolution with holes by replacing the representative disk with a representative **washer**.

The washer is formed by revolving a rectangle about an axis, as shown in Figure 7.18.

If  $r$  and  $R$  are the inner and outer radii of the washer, respectively, and  $w$  is the width of the washer, then the volume is

$$\text{Volume of washer} = \pi(R^2 - r^2)w.$$

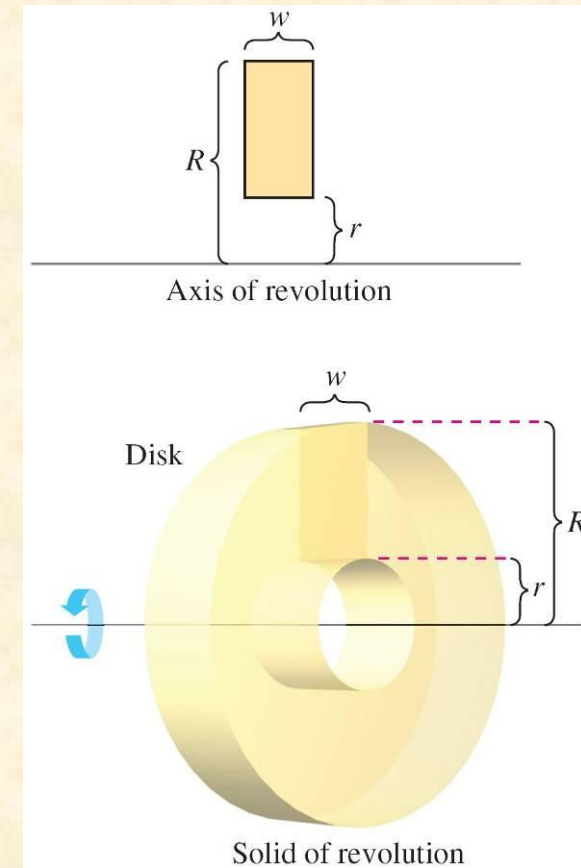


Figure 7.18

# The Washer Method

To see how this concept can be used to find the volume of a solid of revolution, consider a region bounded by an **outer radius**  $R(x)$  and an **inner radius**  $r(x)$ , as shown in Figure 7.19.

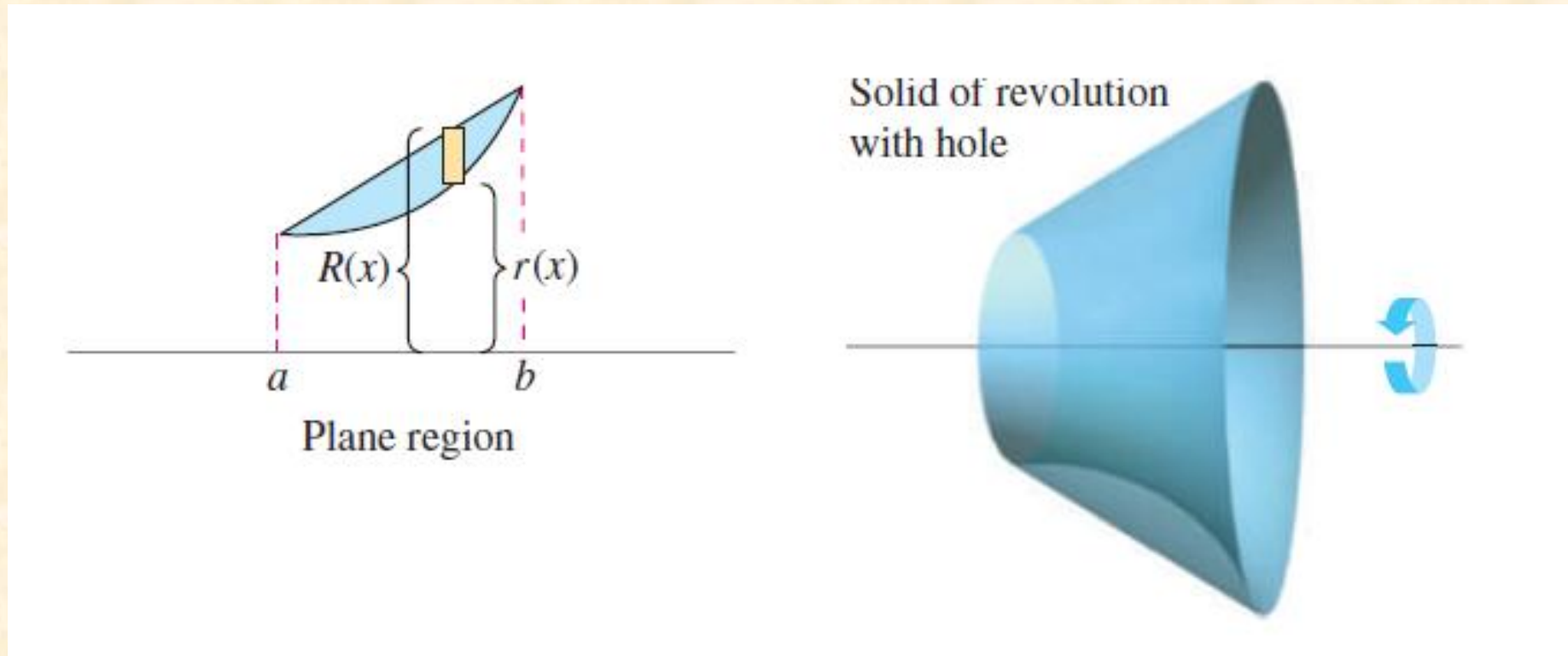


Figure 7.19

# The Washer Method

If the region is revolved about its axis of revolution, then the volume of the resulting solid is

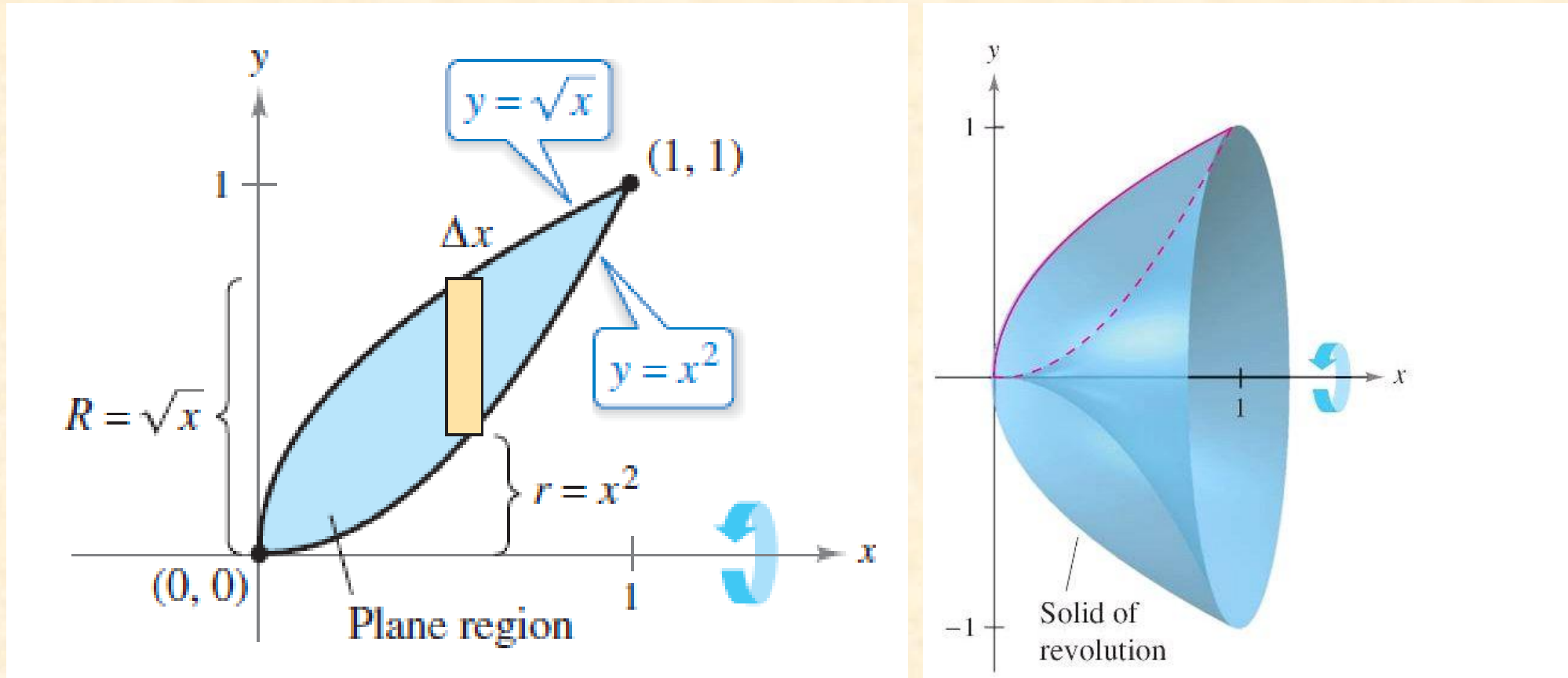
$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx.$$

Washer method

Note that the integral involving the inner radius represents the volume of the hole and is *subtracted* from the integral involving the outer radius.

## Example 3 – Using the Washer Method

Find the volume of the solid formed by revolving the region bounded by the graphs of  $y = \sqrt{x}$  and  $y = x^2$  about the  $x$ -axis, as shown in Figure 7.20.



Solid of revolution

Figure 7.20

## Example 3 – Solution

In Figure 7.20, you can see that the outer and inner radii are as follows.

$$R(x) = \sqrt{x}$$

Outer radius

$$r(x) = x^2$$

Inner radius

Integrating between 0 and 1 produces

$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx$$

Apply washer method.

$$= \pi \int_0^1 [(\sqrt{x})^2 - (x^2)^2] dx$$

Substitute  $\sqrt{x}$  for  $R(x)$  and  $x^2$  for  $r(x)$ .

$$= \pi \int_0^1 (x - x^4) dx$$

Simplify.

$$= \pi \left[ \frac{x^2}{2} - \frac{x^5}{5} \right]_0^1$$

Integrate.

$$= \frac{3\pi}{10}.$$

## Example 4 – Integrating with Respect to $y$ : Two-Integral Case

Find the volume of the solid formed by revolving the region bounded by the graphs of  $y = x^2 + 1$ ,  $y = 0$ ,  $x = 0$ , and  $x = 1$  about  $y$ -axis, as shown in Figure 7.21.

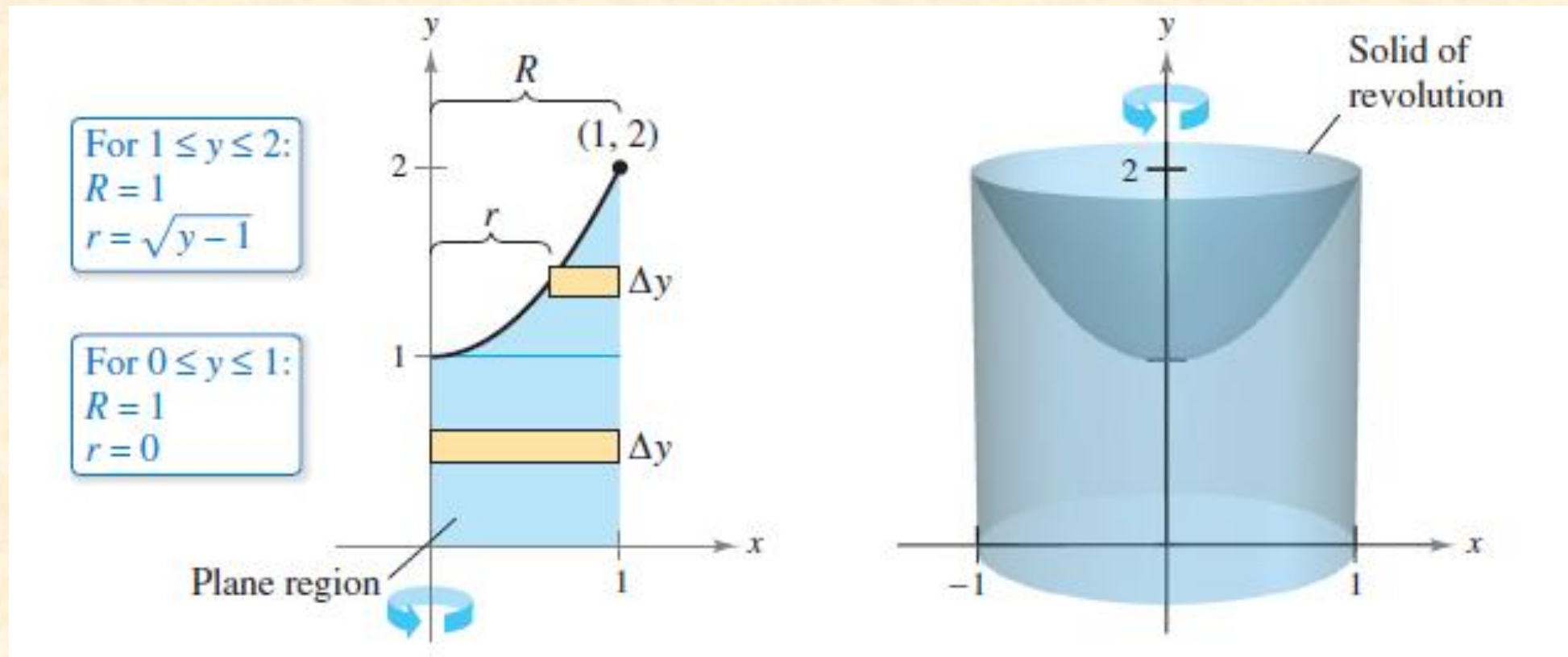


Figure 7.21

## Example 4 – *Solution*

For the region shown in Figure 7.21, the outer radius is simply  $R = 1$ .

There is, however, no convenient formula that represents the inner radius.

When  $0 \leq y \leq 1$ ,  $r = 0$ , but when  $1 \leq y \leq 2$ ,  $r$  is determined by the equation  $y = x^2 + 1$ , which implies that

$$r = \sqrt{y - 1}.$$

$$r(y) = \begin{cases} 0, & 0 \leq y \leq 1 \\ \sqrt{y - 1}, & 1 \leq y \leq 2 \end{cases}$$

## Example 4 – *Solution*

Using this definition of the inner radius, you can use two integrals to find the volume.

$$V = \pi \int_0^1 (1^2 - 0^2) dy + \pi \int_1^2 [1^2 - (\sqrt{y-1})^2] dy$$

Apply washer method.

$$= \pi \int_0^1 1 dy + \pi \int_1^2 (2 - y) dy$$

Simplify.

$$= \pi \left[ y \right]_0^1 + \pi \left[ 2y - \frac{y^2}{2} \right]_1^2$$

Integrate.

$$= \pi + \pi \left( 4 - 2 - 2 + \frac{1}{2} \right)$$

$$= \frac{3\pi}{2}$$

# Solids with Known Cross Sections

- With the disk method, you can find the volume of a solid having a circular cross section whose area is  $A = \pi R^2$ .
- This method can be generalized to solids of any shape, as long as you know a formula for the area of an arbitrary cross section.
- Some common cross sections are squares, rectangles, triangles, semicircles, and trapezoids.

# Solids with Known Cross Sections

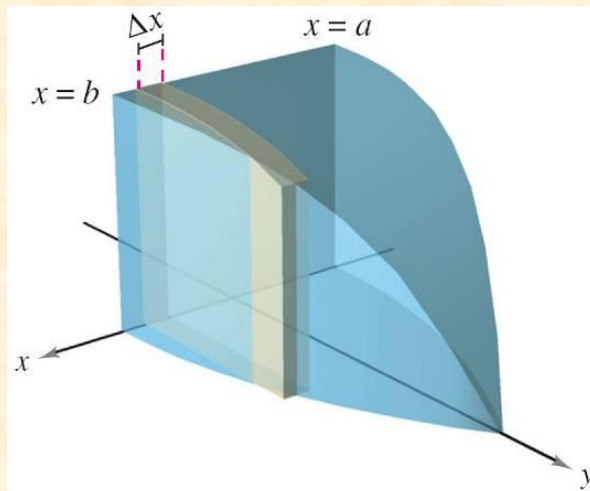
## VOLUMES OF SOLIDS WITH KNOWN CROSS SECTIONS

1. For cross sections of area  $A(x)$  taken perpendicular to the  $x$ -axis,

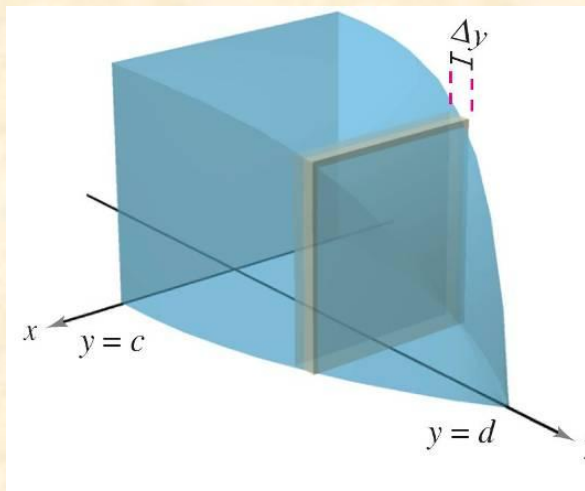
$$\text{Volume} = \int_a^b A(x) dx. \quad \text{See Figure 7.24(a).}$$

2. For cross sections of area  $A(y)$  taken perpendicular to the  $y$ -axis,

$$\text{Volume} = \int_c^d A(y) dy. \quad \text{See Figure 7.24(b).}$$



(a) Cross sections perpendicular to  $x$ -axis



(b) Cross sections perpendicular to  $y$ -axis

Figure 7.24

## Example 6 – Triangular Cross Sections

Find the volume of the solid shown in Figure 7.25. The base of the solid is the region bounded by the lines  $f(x) = 1 - \frac{x}{2}$ ,  $g(x) = -1 + \frac{x}{2}$ , and  $x = 0$ .

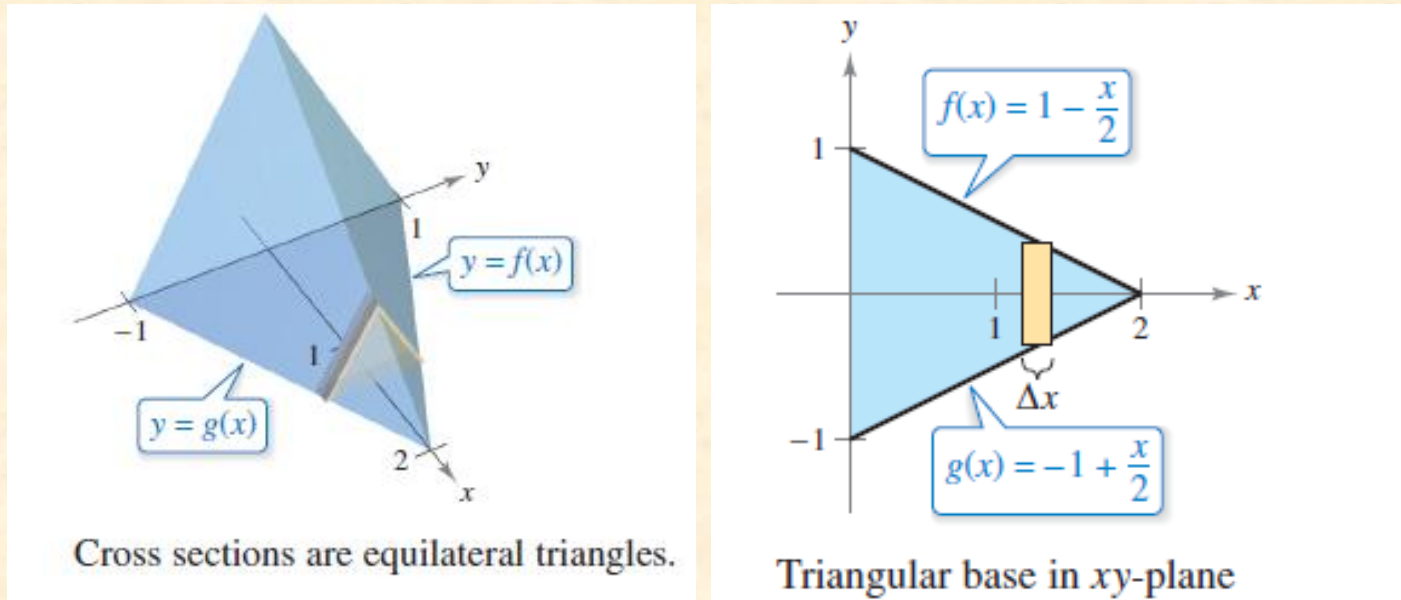


Figure 7.25

The cross sections perpendicular to the  $x$ -axis are equilateral triangles.

## Example 6 – Solution

The base and area of each triangular cross section are as follows.

$$\text{Base} = \left(1 - \frac{x}{2}\right) - \left(-1 + \frac{x}{2}\right) = 2 - x \quad \text{Length of base}$$

$$\text{Area} = \frac{\sqrt{3}}{4} (\text{base})^2 \quad \text{Area of equilateral triangle}$$

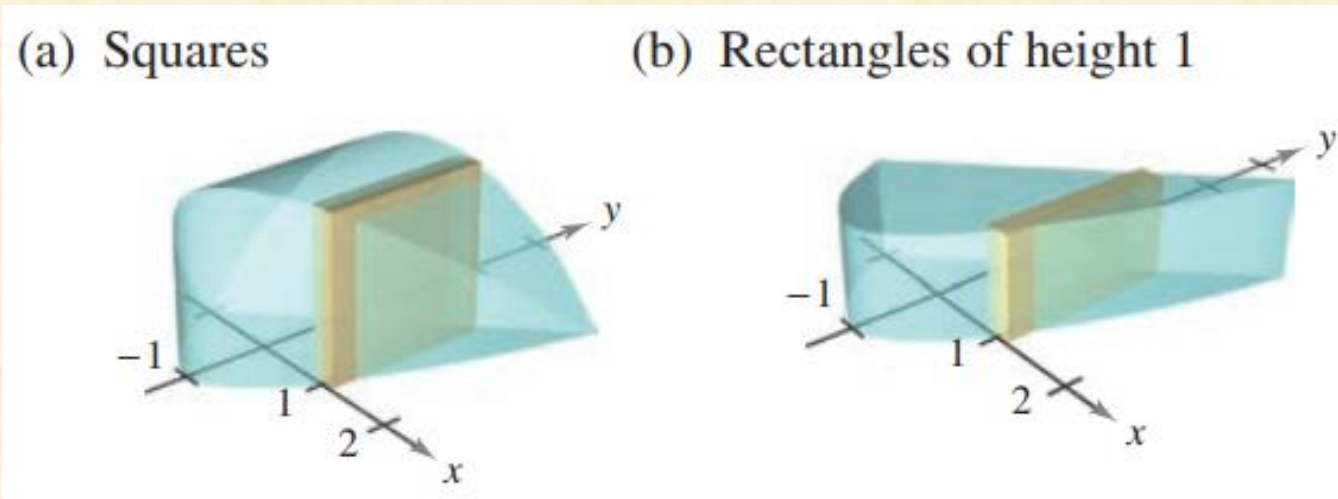
$$A(x) = \frac{\sqrt{3}}{4} (2 - x)^2 \quad \text{Area of cross section}$$

Because  $x$  ranges from 0 to 2, the volume of the solid is

$$V = \int_a^b A(x) \, dx = \int_0^2 \frac{\sqrt{3}}{4} (2 - x)^2 \, dx = -\frac{\sqrt{3}}{4} \left[ \frac{(2 - x)^3}{3} \right]_0^2 = \frac{2\sqrt{3}}{3}.$$

# Try Yourself

1. Find the volumes of the solids whose bases are bounded by the graphs of  $y = x + 1$  and  $y = x^2 - 1$ , with the indicated cross sections taken perpendicular to the  $x$ -axis.



## **7.3 Volume: The Shell Method**

# The Shell Method

An alternative method for finding the volume of a solid of revolution is called the **shell method** because it uses cylindrical shells.

A comparison of the advantages of the disk and shell methods is given later in this section.

Consider a representative rectangle as shown in Figure 7.27, where  $w$  is the width of the rectangle,  $h$  is the height of the rectangle, and  $p$  is the distance between the axis of revolution and the *center* of the rectangle.

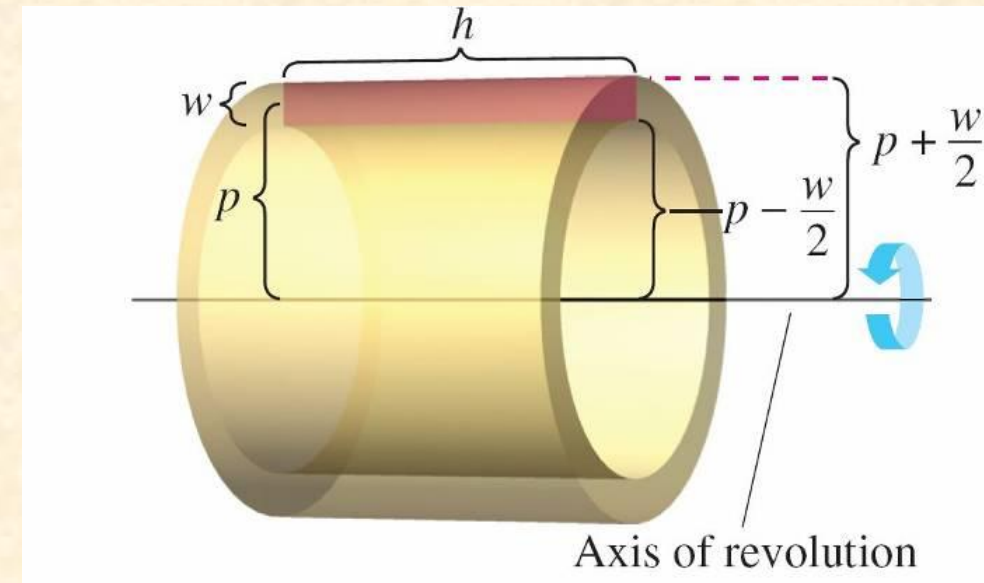


Figure 7.27

# The Shell Method

When this rectangle is revolved about its axis of revolution, it forms a cylindrical shell (or tube) of thickness  $w$ .

To find the volume of this shell, consider two cylinders. The radius of the larger cylinder corresponds to the outer radius of the shell, and the radius of the smaller cylinder corresponds to the inner radius of the shell. Because  $p$  is the average radius of the shell, you know the outer radius is

$$p + \frac{w}{2} \quad \text{Outer radius}$$

and the inner radius is

$$p - \frac{w}{2} \quad \text{Inner radius}$$

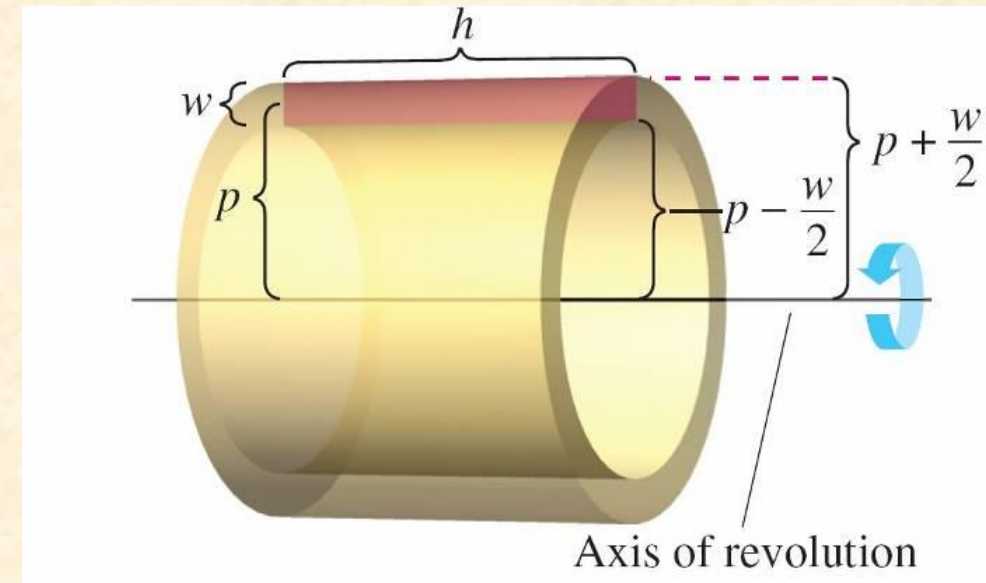


Figure 7.27

# The Shell Method

So, the volume of the shell is

$$\begin{aligned}\text{Volume of shell} &= (\text{volume of cylinder}) - (\text{volume of hole}) \\ &= \pi \left( p + \frac{w}{2} \right)^2 h - \pi \left( p - \frac{w}{2} \right)^2 h \\ &= 2\pi p h w \\ &= 2\pi (\text{average radius})(\text{height})(\text{thickness}).\end{aligned}$$

You can use this formula to find the volume of a solid of revolution.

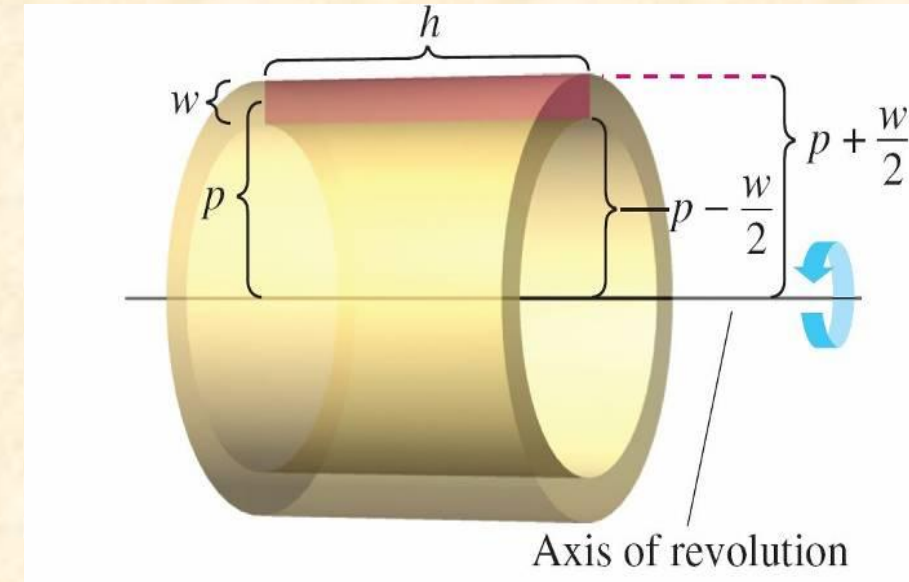


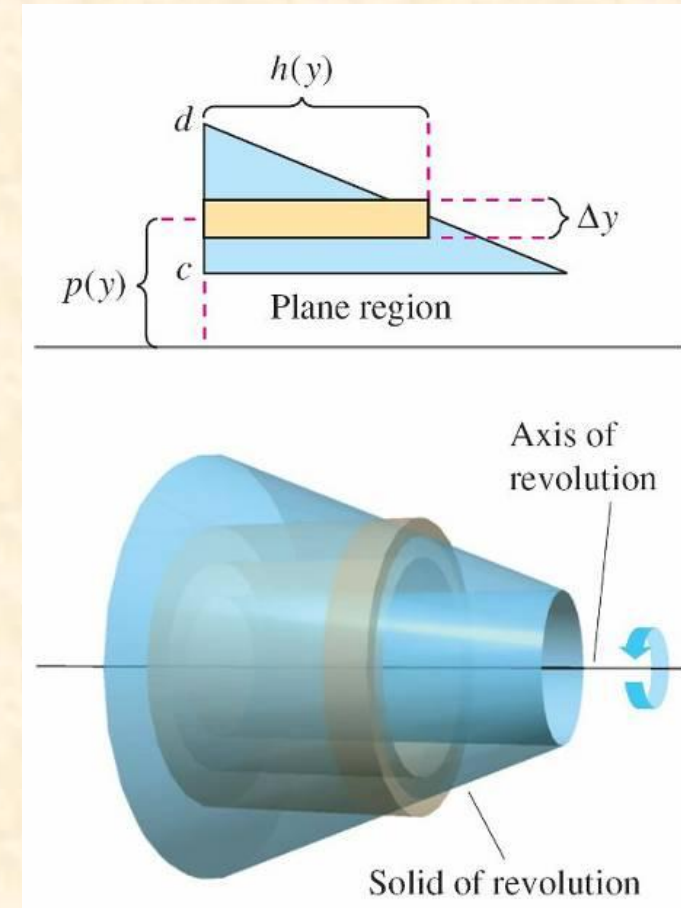
Figure 7.27

# The Shell Method

For instance, the plane region in Figure 7.28 is revolved about a line to form the indicated solid.

Consider a horizontal rectangle of width  $\Delta y$ , then, as the plane region is revolved about a line parallel to the  $x$ -axis, the rectangle generates a representative shell whose volume is

$$\Delta V = 2\pi[p(y)h(y)]\Delta y.$$



**Figure 7.28**

# The Shell Method

You can approximate the volume of the solid by  $n$  such shells of thickness  $\Delta y$ , height  $h(y_i)$ , and average radius  $p(y_i)$ .

$$\text{Volume of solid} \approx \sum_{i=1}^n 2\pi[p(y_i)h(y_i)] \Delta y = 2\pi \sum_{i=1}^n [p(y_i)h(y_i)] \Delta y$$

This approximation appears to become better and better as  $\|\Delta\| \rightarrow 0$  ( $n \rightarrow \infty$ ). So, the volume of the solid is

$$\begin{aligned} \text{Volume of solid} &= \lim_{\|\Delta\| \rightarrow 0} 2\pi \sum_{i=1}^n [p(y_i)h(y_i)] \Delta y \\ &= 2\pi \int_c^d [p(y)h(y)] dy. \end{aligned}$$

# The Shell Method

## THE SHELL METHOD

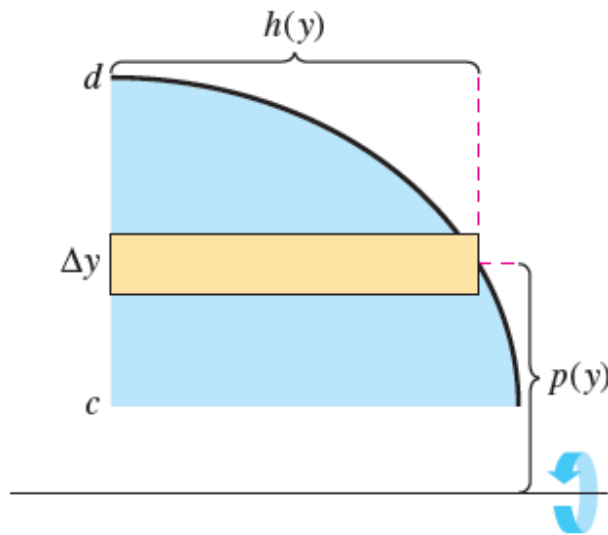
To find the volume of a solid of revolution with the **shell method**, use one of the formulas below. (See Figure 7.29.)

### Horizontal Axis of Revolution

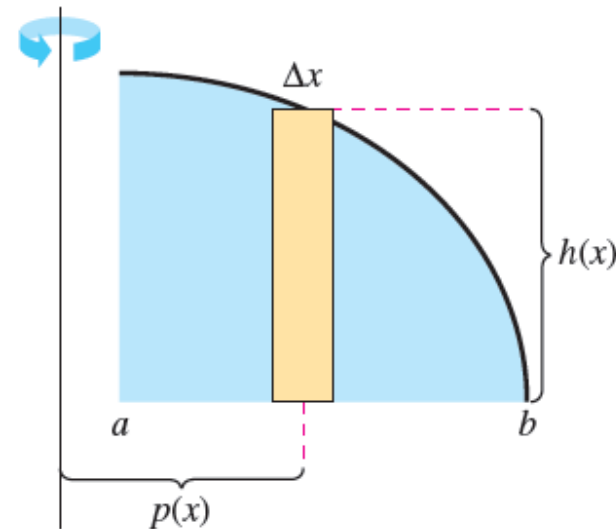
$$\text{Volume} = V = 2\pi \int_c^d p(y)h(y) dy$$

### Vertical Axis of Revolution

$$\text{Volume} = V = 2\pi \int_a^b p(x)h(x) dx$$



Horizontal axis of revolution



Vertical axis of revolution

Figure 7.29

## Example 7 – Using the Shell Method to Find Volume

Find the volume of the solid formed by revolving the region bounded by  $y = x - x^3$  and the  $x$ -axis ( $0 \leq x \leq 1$ ) about the  $y$ -axis.

### Solution:

Because the axis of revolution is vertical, use a vertical representative rectangle, as shown in Figure 7.30.

The width  $\Delta x$  indicates that  $x$  is the variable of integration.

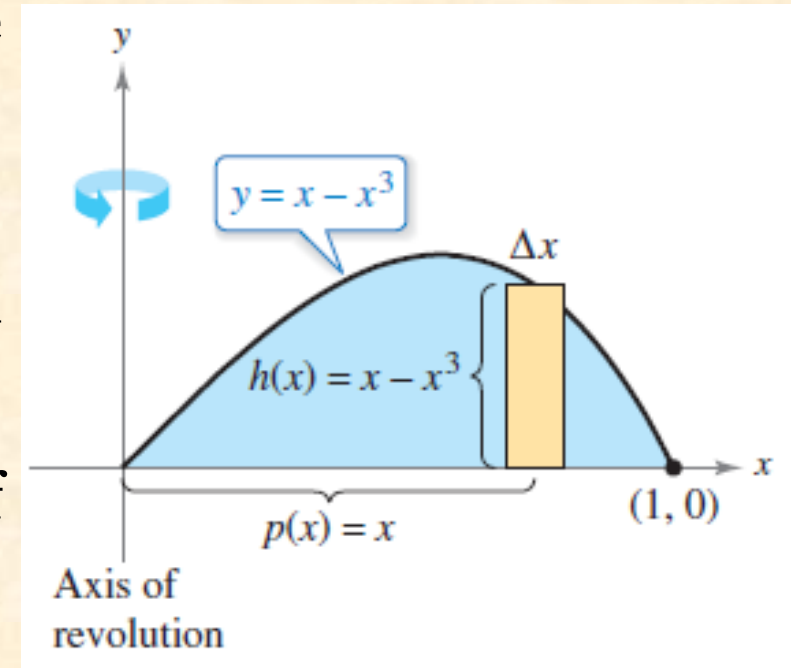


Figure 7.30

## Example 7 – Solution

The distance from the center of the rectangle to the axis of revolution is  $p(x) = x$ , and the height of the rectangle is

$$h(x) = x - x^3.$$

Because  $x$  ranges from 0 to 1, the volume of the solid is

$$V = 2\pi \int_a^b p(x)h(x) dx$$

$$= 2\pi \int_0^1 x(x - x^3) dx$$

$$= 2\pi \int_0^1 (-x^4 + x^2) dx$$

Simplify.

$$= 2\pi \left[ -\frac{x^5}{5} + \frac{x^3}{3} \right]_0^1$$

Integrate.

$$= 2\pi \left( -\frac{1}{5} + \frac{1}{3} \right)$$

$$= \frac{4\pi}{15}$$

# Comparison of Disk and Shell Methods

The disk and shell methods can be distinguished as follows.

For the disk method, the representative rectangle is always *perpendicular* to the axis of revolution, whereas for the shell method, the representative rectangle is always *parallel* to the axis of revolution, as shown in Figure 7.32.

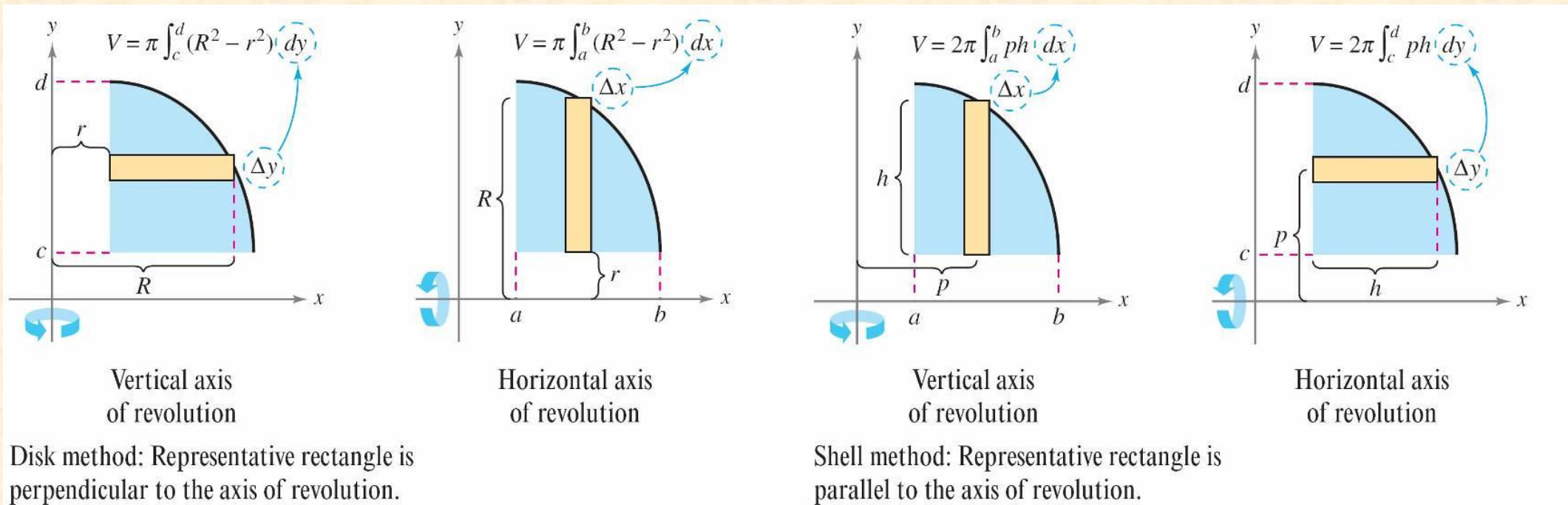


Figure 7.32

## Example 8 – Shell Method Preferable

Find the volume of the solid formed by revolving the region bounded by the graphs of  $y = x^2 + 1$ ,  $y = 0$ ,  $x = 0$ , and  $x = 1$  about the  $y$ -axis.

### Solution:

The washer method requires two integrals to determine the volume of this solid. See Figure 7.33(a).

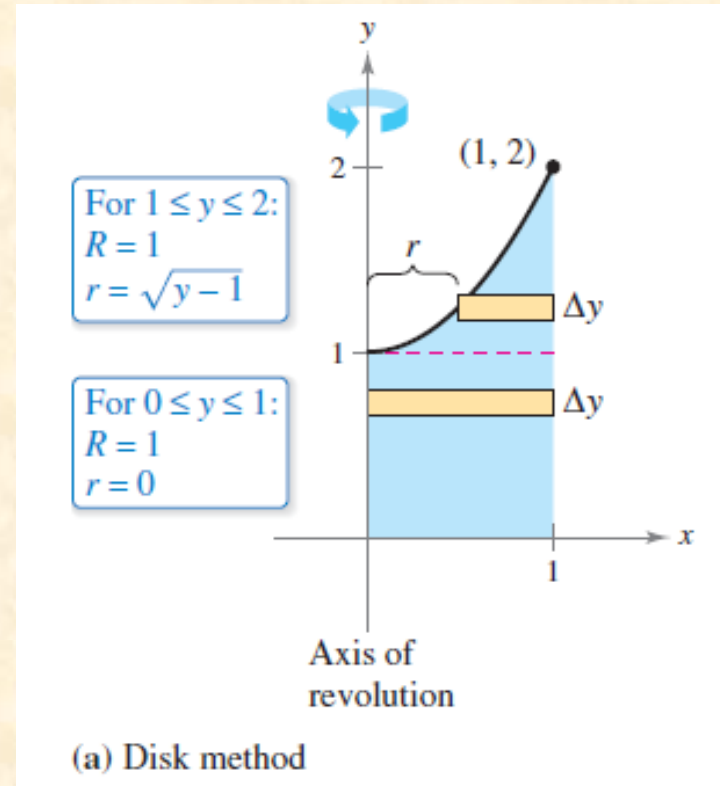


Figure 7.33

## Example 8 – Solution

$$V = \pi \int_0^1 (1^2 - 0^2) dy + \pi \int_1^2 [1^2 - (\sqrt{y-1})^2] dy$$

Apply washer method.

$$= \pi \int_0^1 1 dy + \pi \int_1^2 (2 - y) dy$$

Simplify.

$$= \pi \left[ y \right]_0^1 + \pi \left[ 2y - \frac{y^2}{2} \right]_1^2$$

Integrate.

$$= \pi + \pi \left( 4 - 2 - 2 + \frac{1}{2} \right)$$

$$= \frac{3\pi}{2}$$

## Example 8 – Solution

In Figure 7.33(b), you can see that the shell method requires only one integral to find the volume.

$$V = 2\pi \int_a^b p(x)h(x) dx$$

Apply shell method.

$$= 2\pi \int_0^1 x(x^2 + 1) dx$$

$$= 2\pi \left[ \frac{x^4}{4} + \frac{x^2}{2} \right]_0^1$$

Integrate.

$$= 2\pi \left( \frac{3}{4} \right)$$

$$= \frac{3\pi}{2}$$

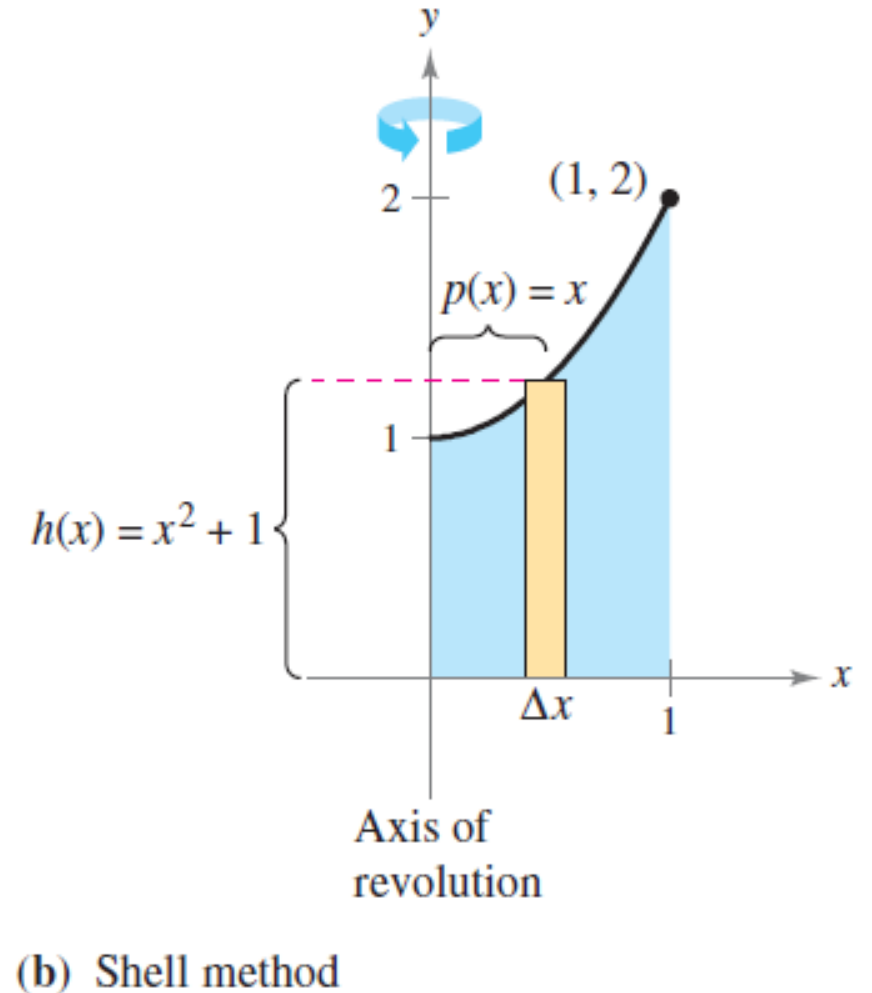


Figure 7.33

## Try Yourself

1. A pond is approximately circular, with a diameter of 400 feet. Starting at the center, the depth of the water is measured every 25 feet and recorded in the table (see figure).

(a) Use the regression capabilities of a graphing utility to find a quadratic model for the depths recorded in the table. Use the graphing utility to plot the depths and graph the model.

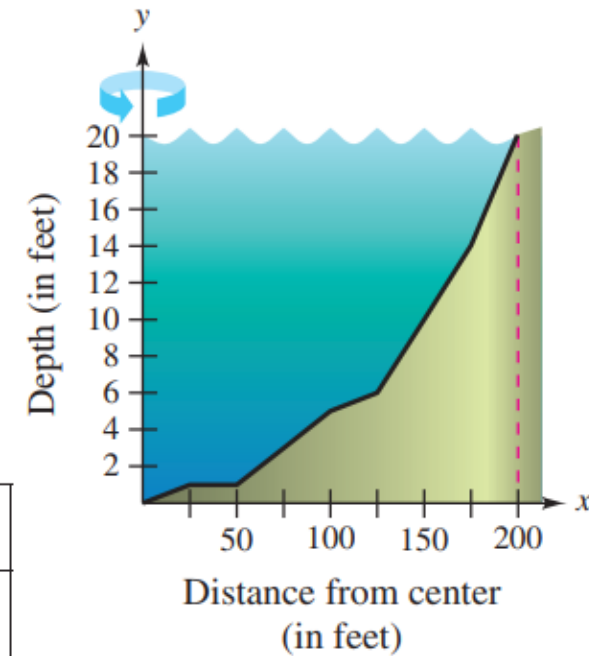
(b) Use the integration capabilities of a graphing utility and the model in part (a) to approximate the volume of water in the pond.

(c) Use the result of part (b) to approximate the number of gallons of water in the pond. (*Hint*: 1 cubic foot of water is approximately 7.48 gallons.)

|       |    |    |    |
|-------|----|----|----|
| $x$   | 0  | 25 | 50 |
| Depth | 20 | 19 | 19 |

|       |    |     |     |
|-------|----|-----|-----|
| $x$   | 75 | 100 | 125 |
| Depth | 17 | 15  | 14  |

|       |     |     |     |
|-------|-----|-----|-----|
| $x$   | 150 | 175 | 200 |
| Depth | 10  | 6   | 0   |



Thanks a lot ...